

Problem 11 Stack

1 • data

2 array db 25h, 35h, 52h, 49h, 53h
3 ; offset address 000F, 0010, 0011, 0012, 0013
4 ; stack offset SP = 0100

5 • code

6 mov ax, array

7 mov ds, ax

8 jmp start

9 MySub Proc uses dx cx ; [Push dx & cx]

10 push bp

11 mov bp, sp

12 sub sp, 4 ; SP = 00F0

13 mov word ptr [bp-4], 59 ; since 59 is single byte forward 0059 = F2

14 mov word ptr [bp-2], 57 ; " 57 [0057] = F2

15 mov dx, [bp-2] ; dx = 0057

16 mov dx, [bp+4] ; dx = 0059

17 mov bx, [bp+10] ; bx = 0012 [bp+10 = FE]

18 mov dx, [bx] ; dx = 53 [0012] = 49 (array)

19 mov sp, bp ; SP = 00F4 b/c word 5349

20 pop bp ; bp = 0000

21 ret 2 ; uses operator so pop cx dx [cx = 49, dx = 4952]

22 MySub ENDP ; SP = 00FF [after pop RA(35) &]
[00FC] cell

23 Start :

24 main Proc

25 mov si, offset array ; si = 000Fh

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Stack Problem (9)

Date:
Sun Mon Tue Wed Thu Fri Sat

10	Fact Proc	(12) SP = BP = 00E8	BP[00EE] (11)	Pop (27)
11	Push bp	(22) SP → 00FA	RA (23)	Pop (27) ← SP
12	mov bp, sp	(21) SP → 00EC	0000(ax)	Pop
13	mov ax, [bp+4]	(18) SP = BP = 00EE	BP(00F4) (11)	Pop (28) ← SP
14	cmp ax, 0	(22) SP 00F0	RA (23)	Pop (28) ← SP
15	Ja L1	(21) SP → 00F2	0001(ax)	Pop (28) ← SP
16	mov ax, 1	(12) SP = BP = 00F4	BP(0000) (11)	Pop (27) ← SP
17	jmp L2	(37) SP → 00F6	RA (38) (main)	(27) ← SP
18	mul bx	00F8	0002(ax)	
19	L1 :	00FA	0000(ax)	
20	dec ax	00FC	0000(dx)	
21	Push ax	(32) SP 00FE	0000(cx)	
22	Call Fact	(30) SP → 0100		
23	return :			
24	mov bx, [bp+4]			
25	mul bx			
26	L2 :	(31)	(36)	(13)
27	pop bp	Ax	0000	0002
28	ret 2	Bx	0001	0002
29	Fact ENDP	Cx	0000	0000
30	main Proc	Dx	0000	0000
31	mov ax, 0	Ax	0001	0002
32	push cx	Bp	0000	00F4
33	push dx		00F4	0000
34	push ax			
35	mov ax, 2			
36	push ax			
37	call Fact			
38	main ENDP			

26 mov word ptr [si], 25 ; [000F] = 25 b/w word 0025, 0000
 27 add si, 2 ; si = 0011
 28 mov dx, [si-2] ; si=000F, [000F] = 25, [0010] = 35, [0011] = 52, [0012] = 49, [0013] = 53
 29 mov dx, [si] ; si=0011, [0011] = 52, [0012] = 49, [0013] = 53
 30 mov di, offset array ; di=000F
 31 add di, 3 ; di=0012
 32 push di
 33 mov cx, 29h ; cx = 49h
 34 push cx
 35 call MySub
 36 inc cx ; cx = 4Ah
 37 push cx
 38 push dx
 39 call AddTwo
 40 mov dx, ax ;
 41 Main ENDP
 42 END main

SP	BP	BP-2	BP	CX	DX	DI
0059	00F0					
0057	00F2					
BP=0000	00F4					
CX: 49	00F6					
RA (40)	00F8					
DX: 4952	00FA					
RA (36) line	00FC					
CX: 4Ah	00FE					
DI: 0012	0100					

After 36 line SP →

43 AddTwo Proc
 44 push BP
 45 mov BP, SP
 46 mov ax, [BP+6] ; BP+6 = 00F6+6 = FC = 50h
 47 mov si, [BP+8] ; BP+8 = FE = 52h, [52] = 49
 48 add ax, [si] ; [0012] = 49 2 word {5349}
 49 pop BP ; BP=0000
 50 ret ; pop RA(40), ax=cx=dx=5399h
 51 AddTwo ENDP

Remaining Stack	dx	cx	di
	4952	50	0012